A: Datasheet

Algorithm: unissey_002

Developer: Unissey

Submission Date: 2024_02_23

Template size: 4096 bytes

Template time (2.5 percentile): 371 msec

Template time (median): 372 msec

Template time (97.5 percentile): 377 msec

Investigation:

Frontal mugshot ranking 444 (out of 481) −− FNIR(1600000, 0, 1) = 0.1810 vs. lowest 0.0008 from intema_001

Mugshot webcam ranking 432 (out of 443) −− FNIR(1600000, 0, 1) = 0.6727 vs. lowest 0.0054 from sensetime_009

Mugshot profile ranking 188 (out of 412) −− FNIR(1600000, 0, 1) = 0.2992 vs. lowest 0.0510 from nec_009

Immigration visa−border ranking 352 (out of 370) −− FNIR(1600000, 0, 1) = 0.6739 vs. lowest 0.0005 from psl_001

Immigration visa−kiosk ranking 310 (out of 314) −− FNIR(1600000, 0, 1) = 0.6982 vs. lowest 0.0387 from cloudwalk_mt_002

Identification:

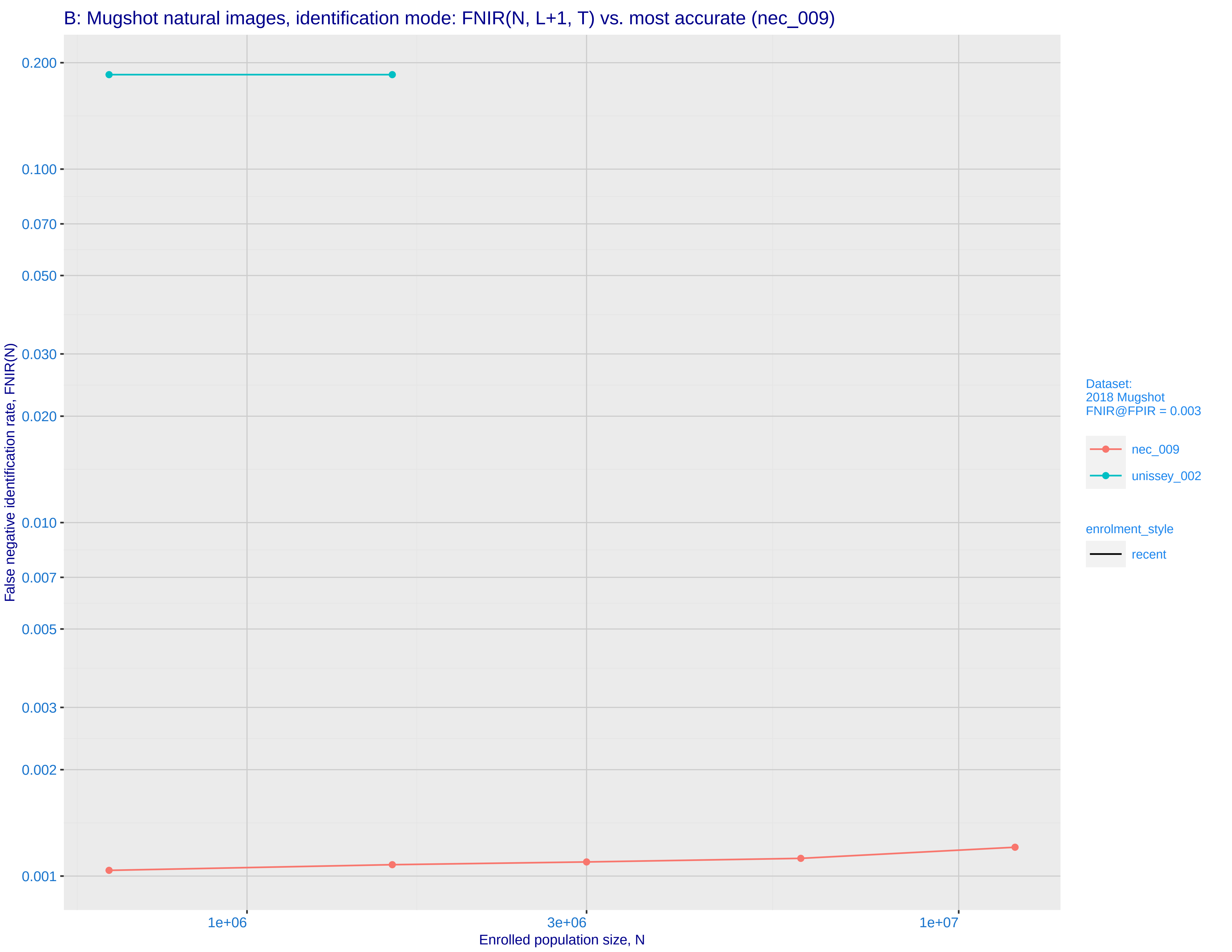
Frontal mugshot ranking 397 (out of 481) −− FNIR(1600000, T, L+1) = 0.1850, FPIR=0.003000 vs. lowest 0.0010 from sensetime_009

Mugshot webcam ranking 412 (out of 441) −− FNIR(1600000, T, L+1) = 0.6800, FPIR=0.003000 vs. lowest 0.0065 from sensetime_009

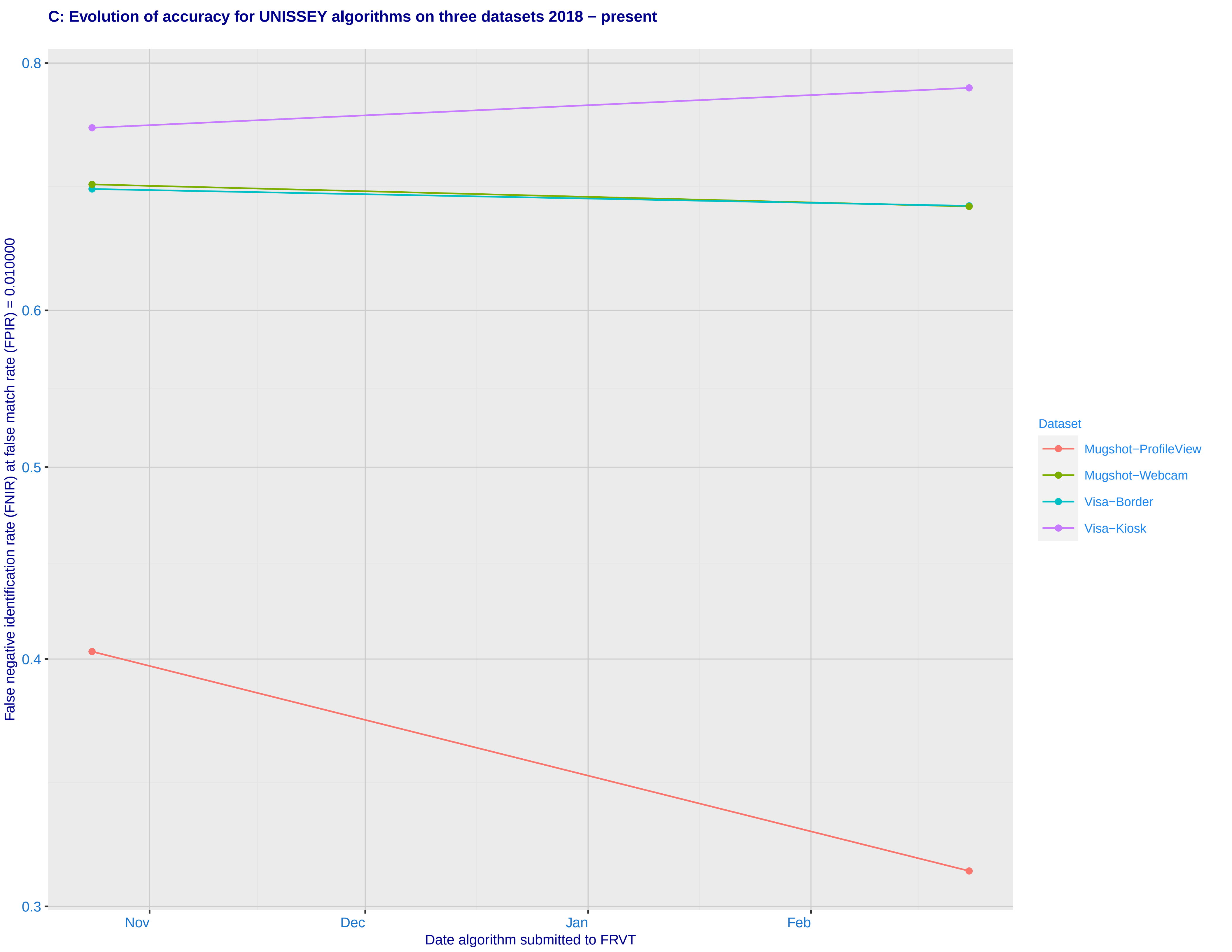
Mugshot profile ranking 108 (out of 411) −− FNIR(1600000, T, L+1) = 0.5029, FPIR=0.003000 vs. lowest 0.0591 from cloudwalk_mt_002

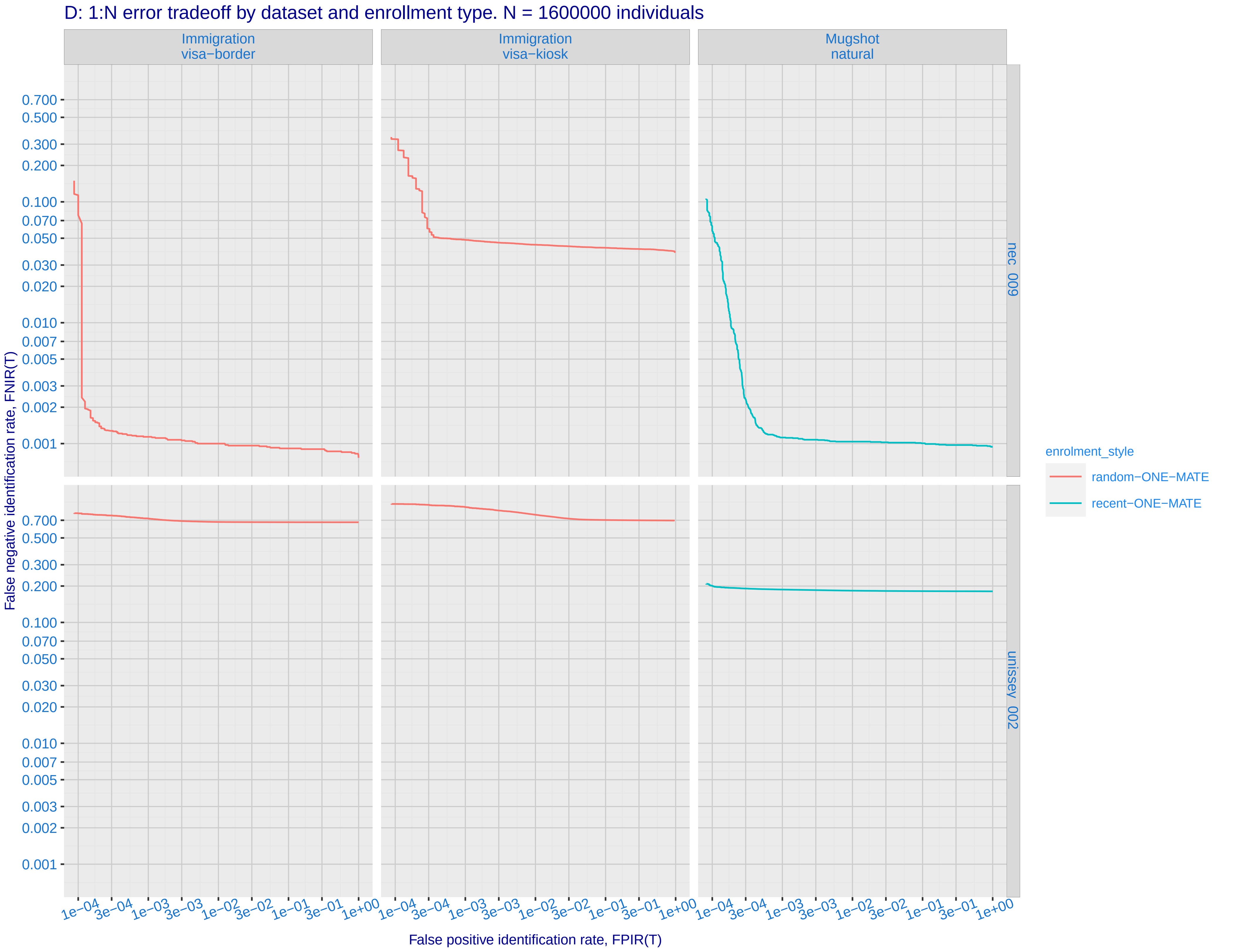
Immigration visa−border ranking 332 (out of 369) −− FNIR(1600000, T, L+1) = 0.6900, FPIR=0.003000 vs. lowest 0.0009 from cloudwalk_mt_002

Immigration visa−kiosk ranking 275 (out of 314) −− FNIR(1600000, T, L+1) = 0.8425, FPIR=0.003000 vs. lowest 0.0460 from nec_009

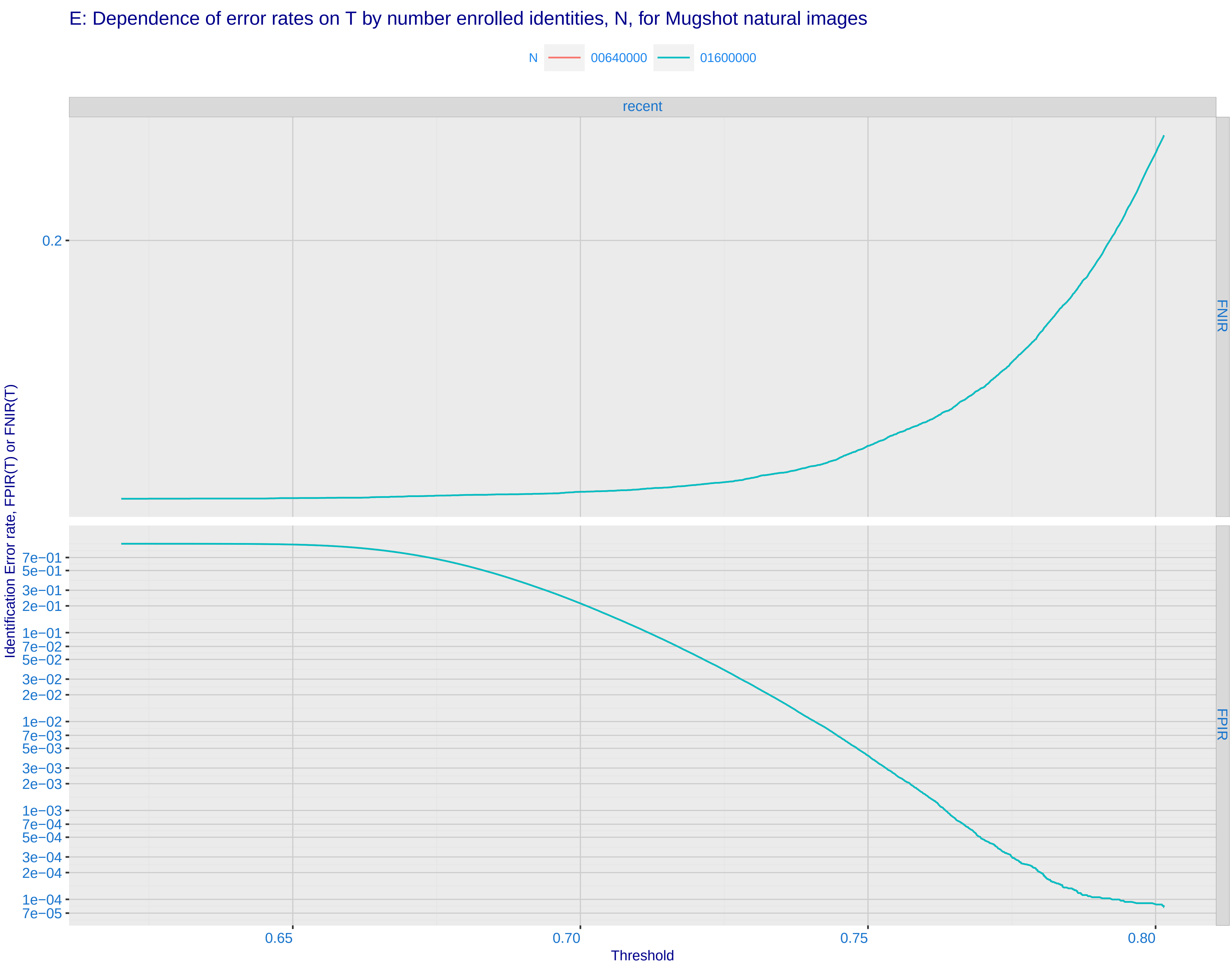


B: Mugshot natural images, identification mode: FNIR(N, L+1, T) vs. most accurate (nec_009)
0.200
0.100
0.070
0.050
0.030
0.020
0.010
0.007
0.005
0.003
0.002
0.001
False negative identification rate, FNIR(N)
1e+06
3e+06
1e+07
Enrolled population size, N
Dataset:
2018 Mugshot
FNIR@FPIR = 0.003
nec_009
unissey_002
enrolment_style
recent

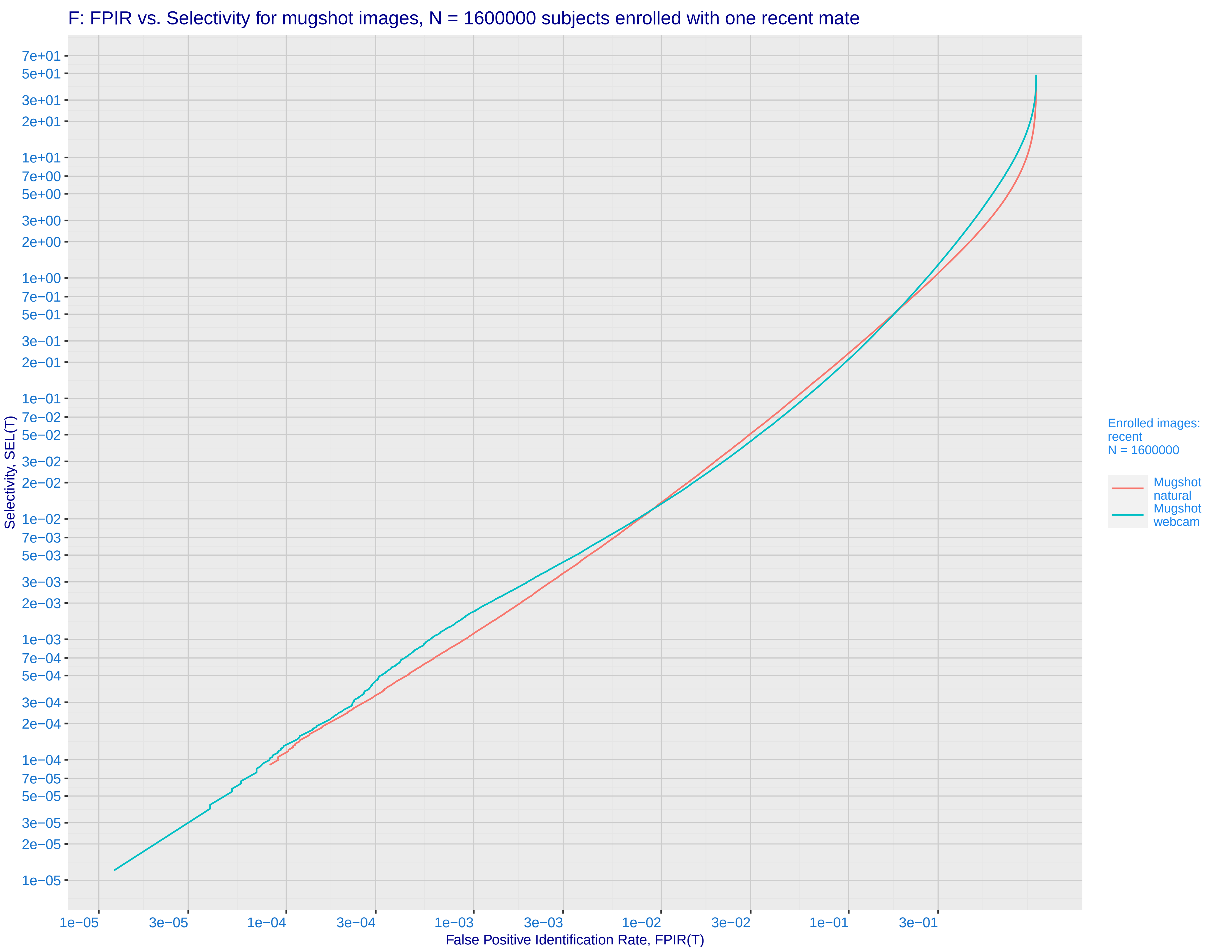

C: Evolution of accuracy for UNISSEY algorithms on three datasets 2018 − present
0.8
0.6
0.5
0.4
0.3
False negative identification rate (FNIR) at false match rate (FPIR) = 0.010000
Nov
Dec
Jan
Feb
Date algorithm submitted to FRVT
Dataset
Mugshot−ProfileView
Mugshot−Webcam
Visa−Border
Visa−Kiosk

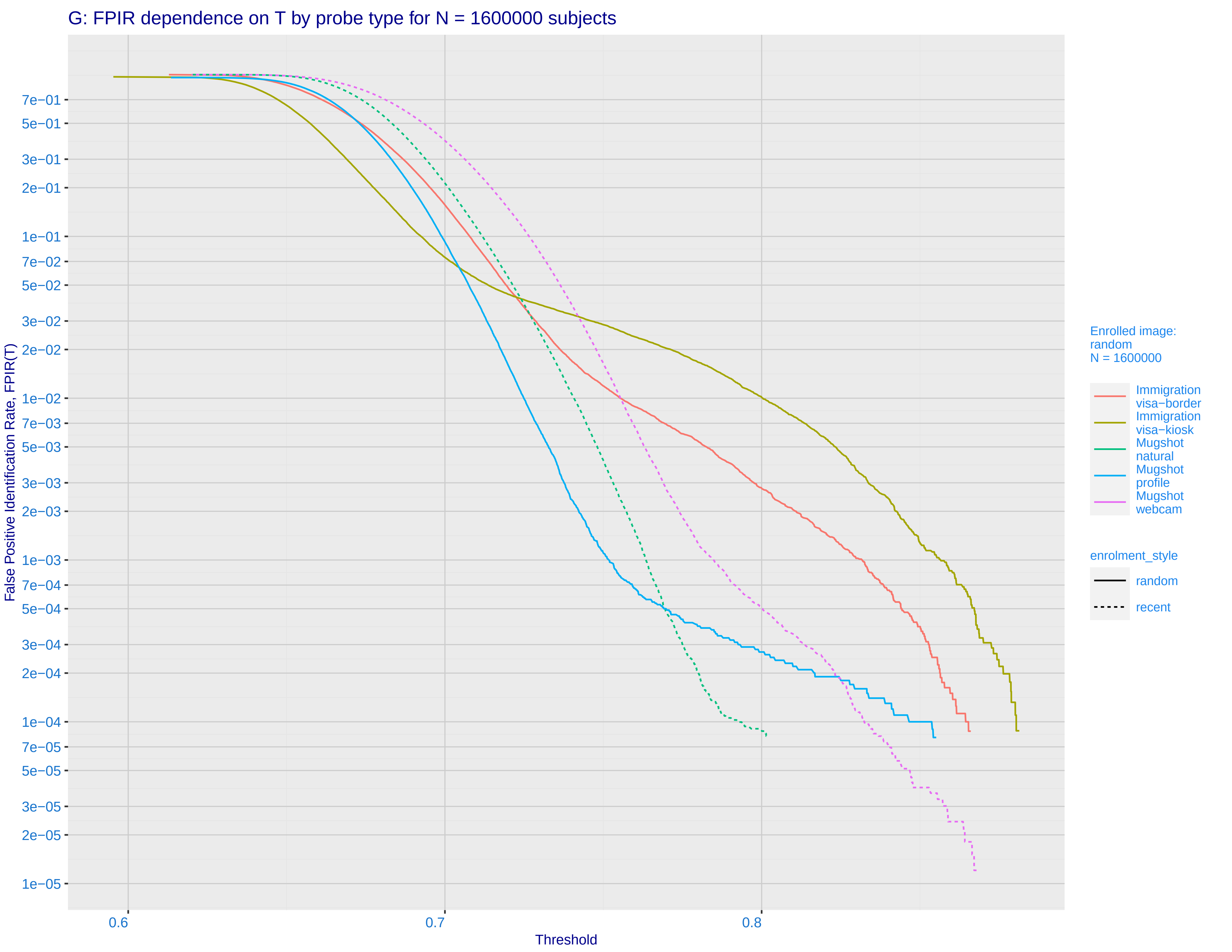


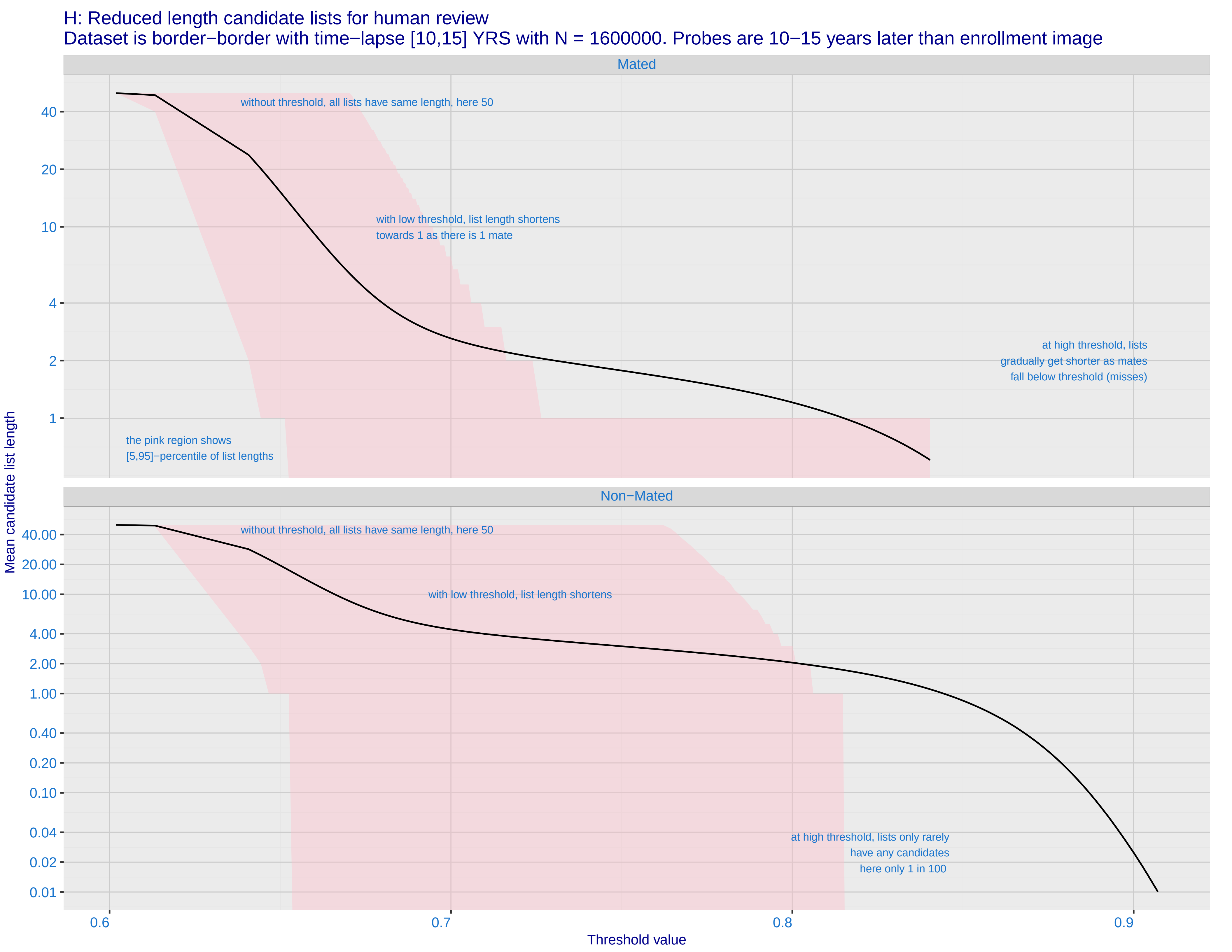
D: 1:N error tradeoff by dataset and enrollment type. N = 1600000 individuals
Immigration
visa−border
Immigration
visa−kiosk
Mugshot
natural
nec_009
unissey_002
False negative identification rate, FNIR(T)
False positive identification rate, FPIR(T)
0.700
0.500
0.300
0.200
0.100
0.070
0.050
0.030
0.020
0.010
0.007
0.005
0.003
0.002
0.001
1e−04
3e−04
1e−03
3e−03
1e−02
3e−02
1e−01
3e−01
1e+00
enrolment_style
random−ONE−MATE
recent−ONE−MATE

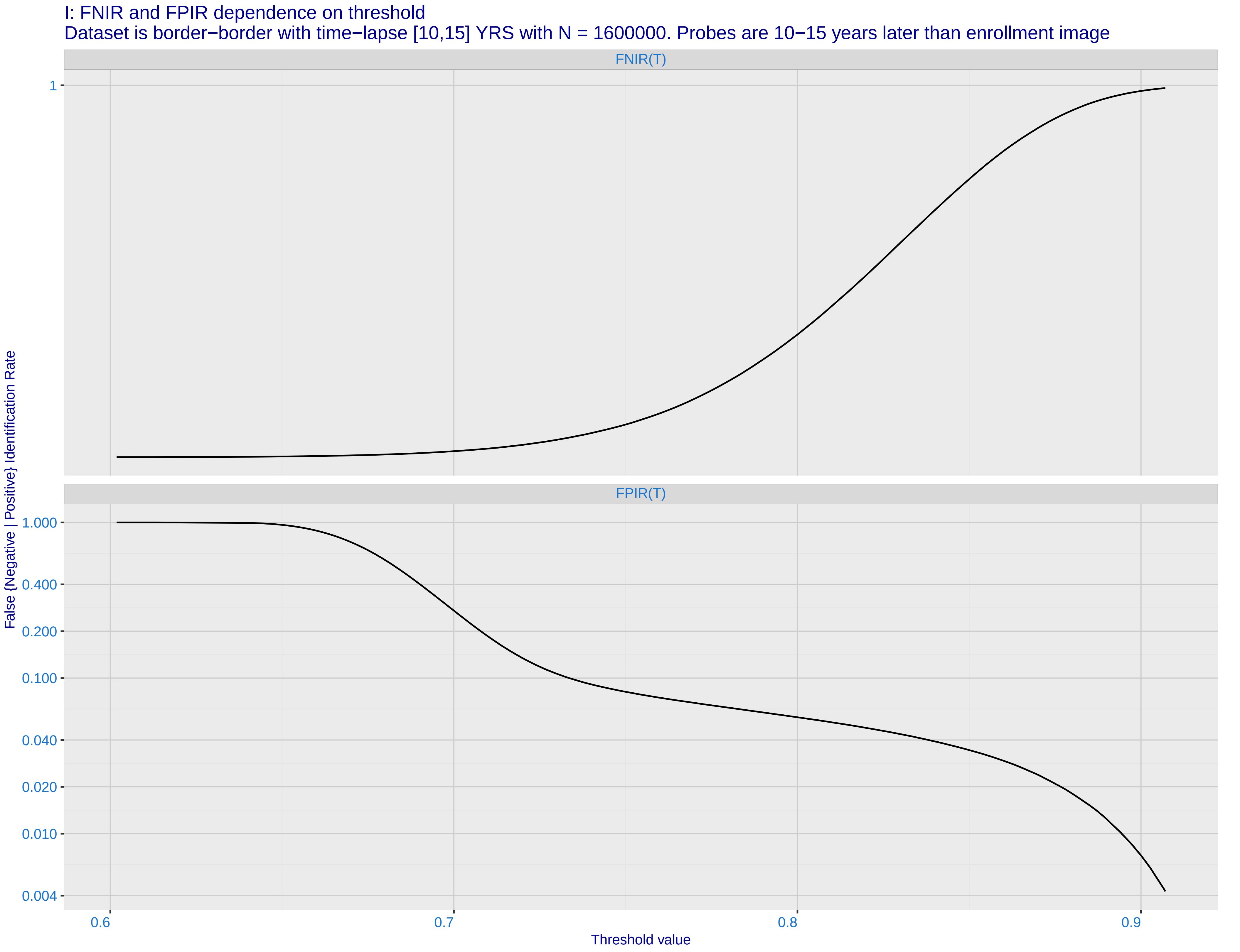


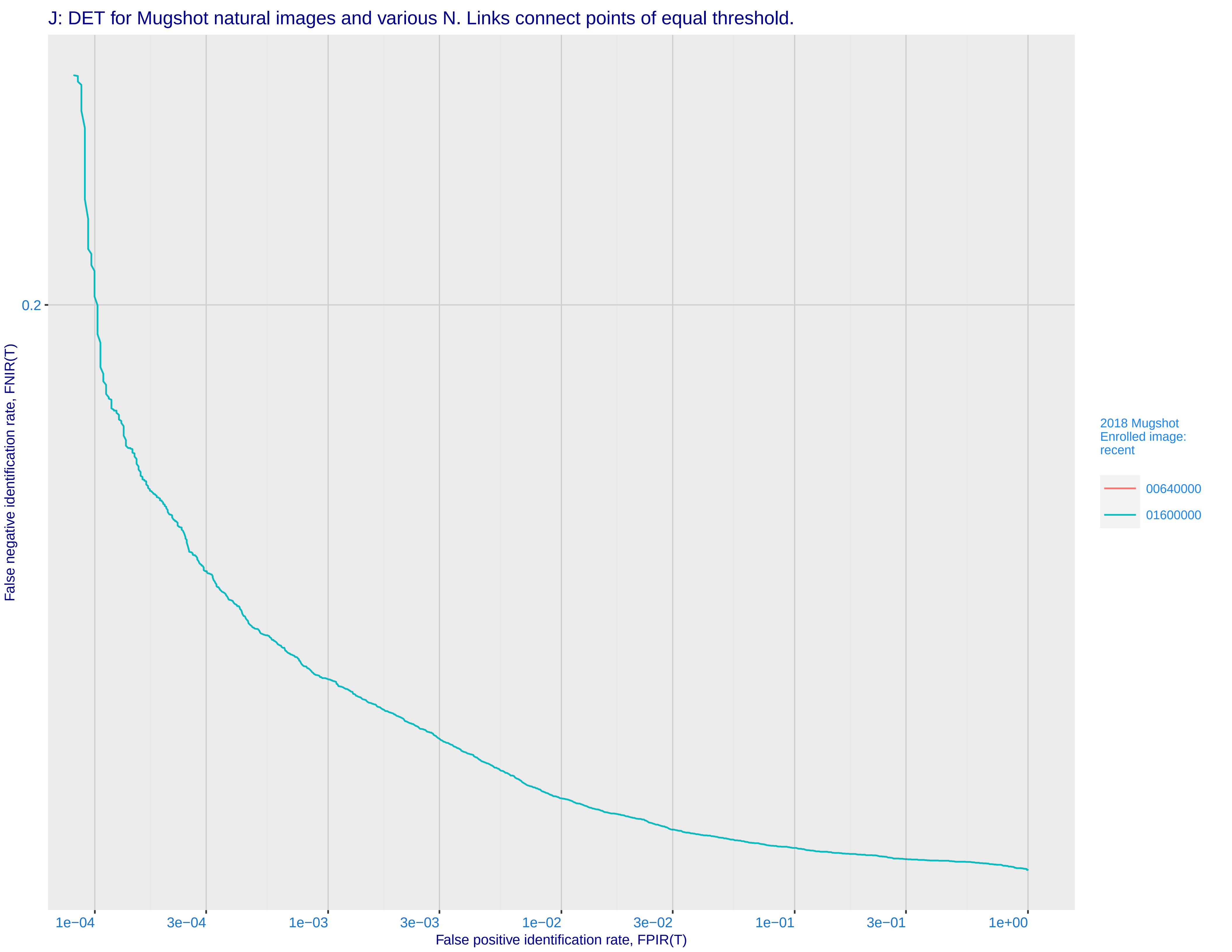
E: Dependence of error rates on T by number enrolled identities, N, for Mugshot natural images
N
00640000
01600000
recent
FNIR
FPIR
0.2
7e−01
5e−01
3e−01
2e−01
1e−01
7e−02
5e−02
3e−02
2e−02
1e−02
7e−03
5e−03
3e−03
2e−03
1e−03
7e−04
5e−04
3e−04
2e−04
1e−04
7e−05
0.65
0.70
0.75
0.80
Threshold
Identification Error rate, FPIR(T) or FNIR(T)


F: FPIR vs. Selectivity for mugshot images, N = 1600000 subjects enrolled with one recent mate
Selectivity, SEL(T)
7e+01
5e+01
3e+01
2e+01
1e+01
7e+00
5e+00
3e+00
2e+00
1e+00
7e−01
5e−01
3e−01
2e−01
1e−01
7e−02
5e−02
3e−02
2e−02
1e−02
7e−03
5e−03
3e−03
2e−03
1e−03
7e−04
5e−04
3e−04
2e−04
1e−04
7e−05
5e−05
3e−05
2e−05
1e−05
1e−05
3e−05
1e−04
3e−04
1e−03
3e−03
1e−02
3e−02
1e−01
3e−01
False Positive Identification Rate, FPIR(T)
Enrolled images:
recent
N = 1600000
Mugshot
natural
Mugshot
webcam

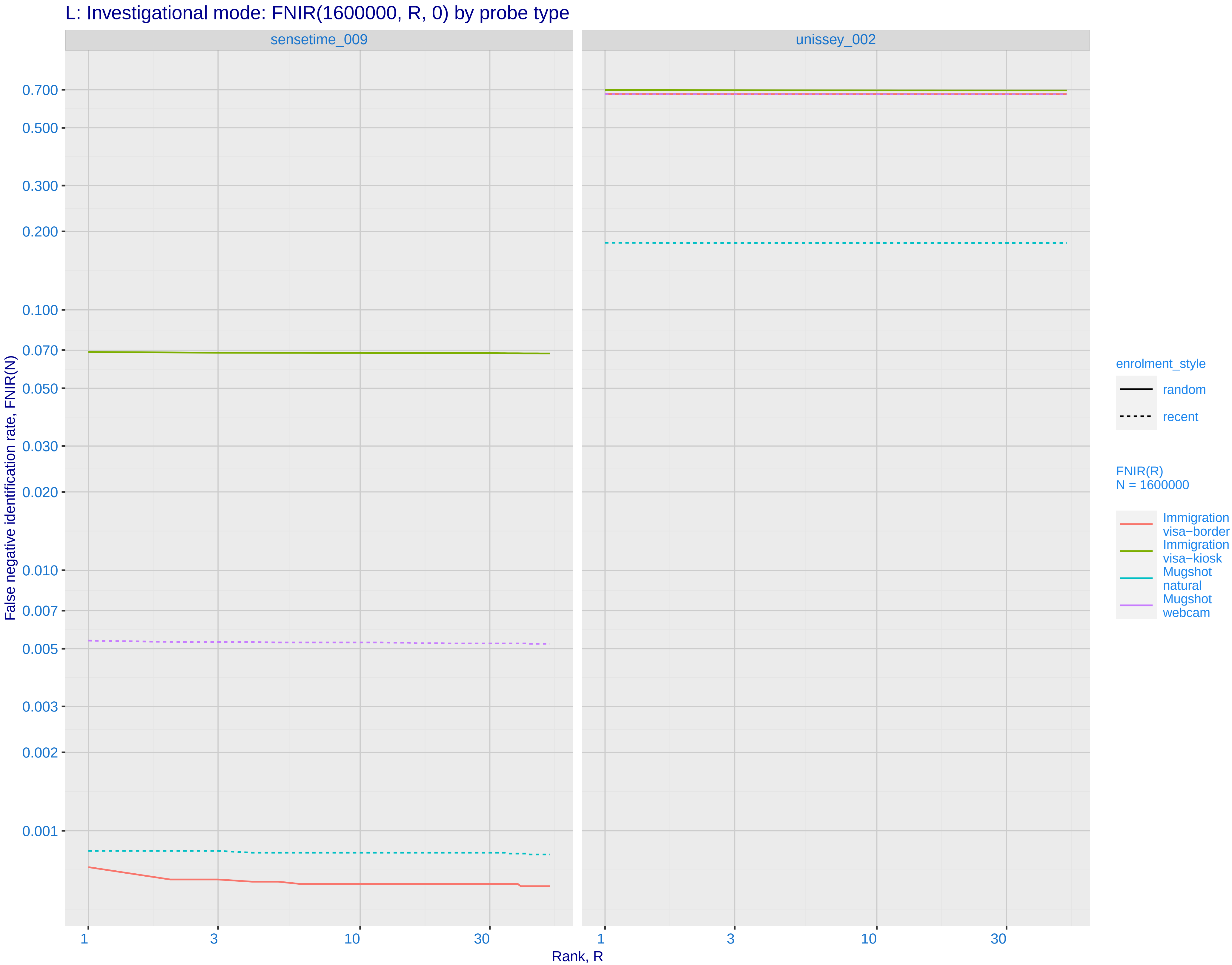

G: FPIR dependence on T by probe type for N = 1600000 subjects
False Positive Identification Rate, FPIR(T)
7e−01
5e−01
3e−01
2e−01
1e−01
7e−02
5e−02
3e−02
2e−02
1e−02
7e−03
5e−03
3e−03
2e−03
1e−03
7e−04
5e−04
3e−04
2e−04
1e−04
7e−05
5e−05
3e−05
2e−05
1e−05
0.6
0.7
0.8
Threshold
Enrolled image:
random
N = 1600000
Immigration visa−border
Immigration visa−kiosk
Mugshot natural
Mugshot profile
Mugshot webcam
enrolment_style
random
recent


H: Reduced length candidate lists for human review
Dataset is border−border with time−lapse [10,15] YRS with N = 1600000. Probes are 10−15 years later than enrollment image
Mated
without threshold, all lists have same length, here 50
with low threshold, list length shortens
towards 1 as there is 1 mate
at high threshold, lists
gradually get shorter as mates
fall below threshold (misses)
the pink region shows
[5,95]−percentile of list lengths
40
20
10
4
2
1
Non−Mated
without threshold, all lists have same length, here 50
with low threshold, list length shortens
at high threshold, lists only rarely
have any candidates
here only 1 in 100
40.00
20.00
10.00
4.00
2.00
1.00
0.40
0.20
0.10
0.04
0.02
0.01
Mean candidate list length
0.6
0.7
0.8
0.9
Threshold value


I: FNIR and FPIR dependence on threshold
Dataset is border−border with time−lapse [10,15] YRS with N = 1600000. Probes are 10−15 years later than enrollment image
FNIR(T)
1
FPIR(T)
1.000
0.400
0.200
0.100
0.040
0.020
0.010
0.004
0.6
0.7
0.8
0.9
False {Negative | Positive} Identification Rate
Threshold value

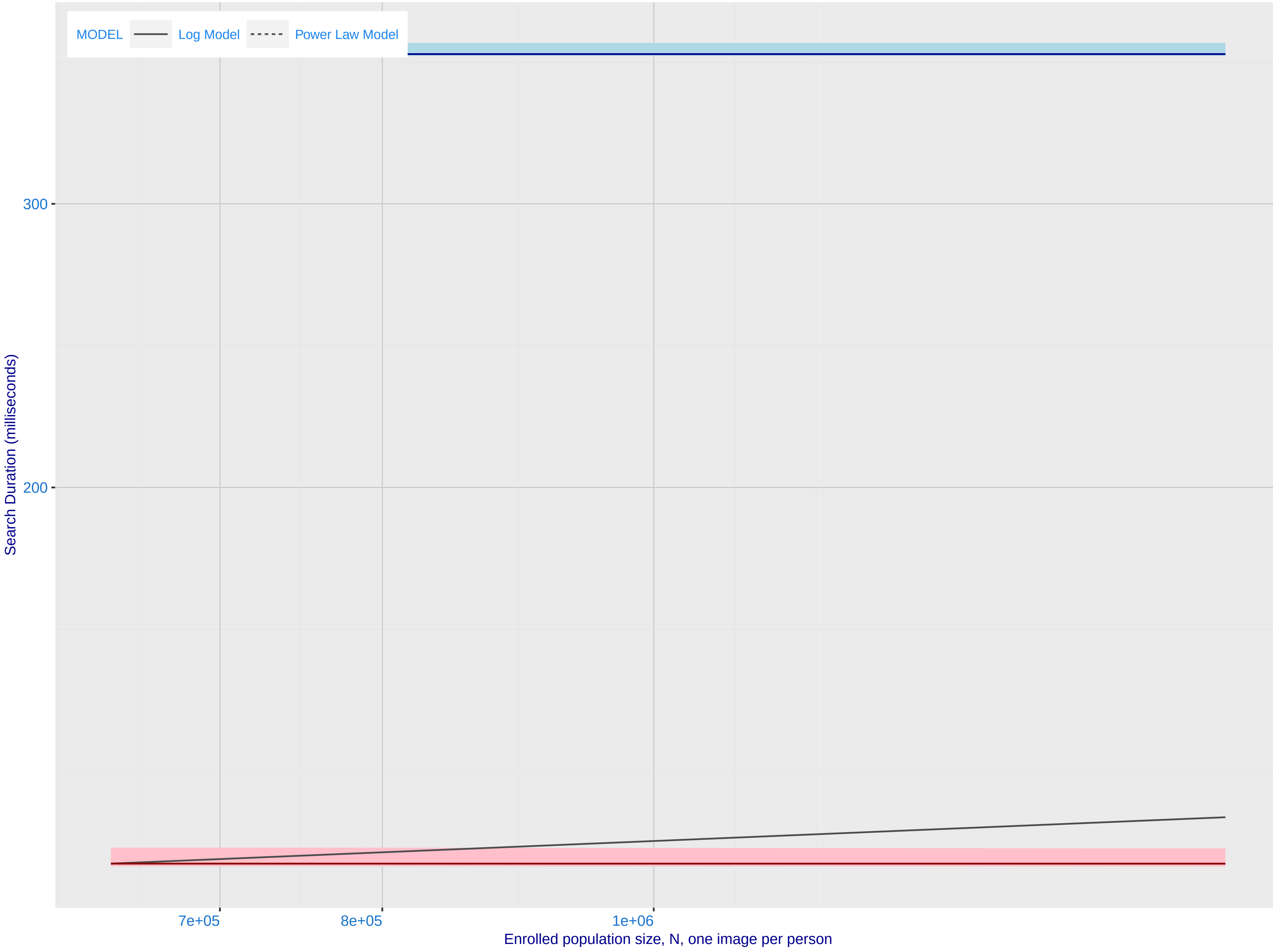

J: DET for Mugshot natural images and various N. Links connect points of equal threshold.
0.2
False negative identification rate, FNIR(T)
1e−04
3e−04
1e−03
3e−03
1e−02
3e−02
1e−01
3e−01
1e+00
False positive identification rate, FPIR(T)
2018 Mugshot
Enrolled image:
recent
00640000
01600000


K: Investigational mode: FNIR(N, 1, 0) vs. most accurate (sensetime_009)
Immigration
visa−border
Immigration
visa−kiosk
Mugshot
natural
Mugshot
webcam
0.700
0.500
0.300
0.200
0.100
0.070
0.050
0.030
0.020
0.010
0.007
0.005
0.003
0.002
0.001
False negative identification rate, FNIR(N)
1e+06
3e+06
1e+07
Enrolled population size, N
FNIR@Rank = 1
sensetime_009
unissey_002
enrolment_style
random
recent

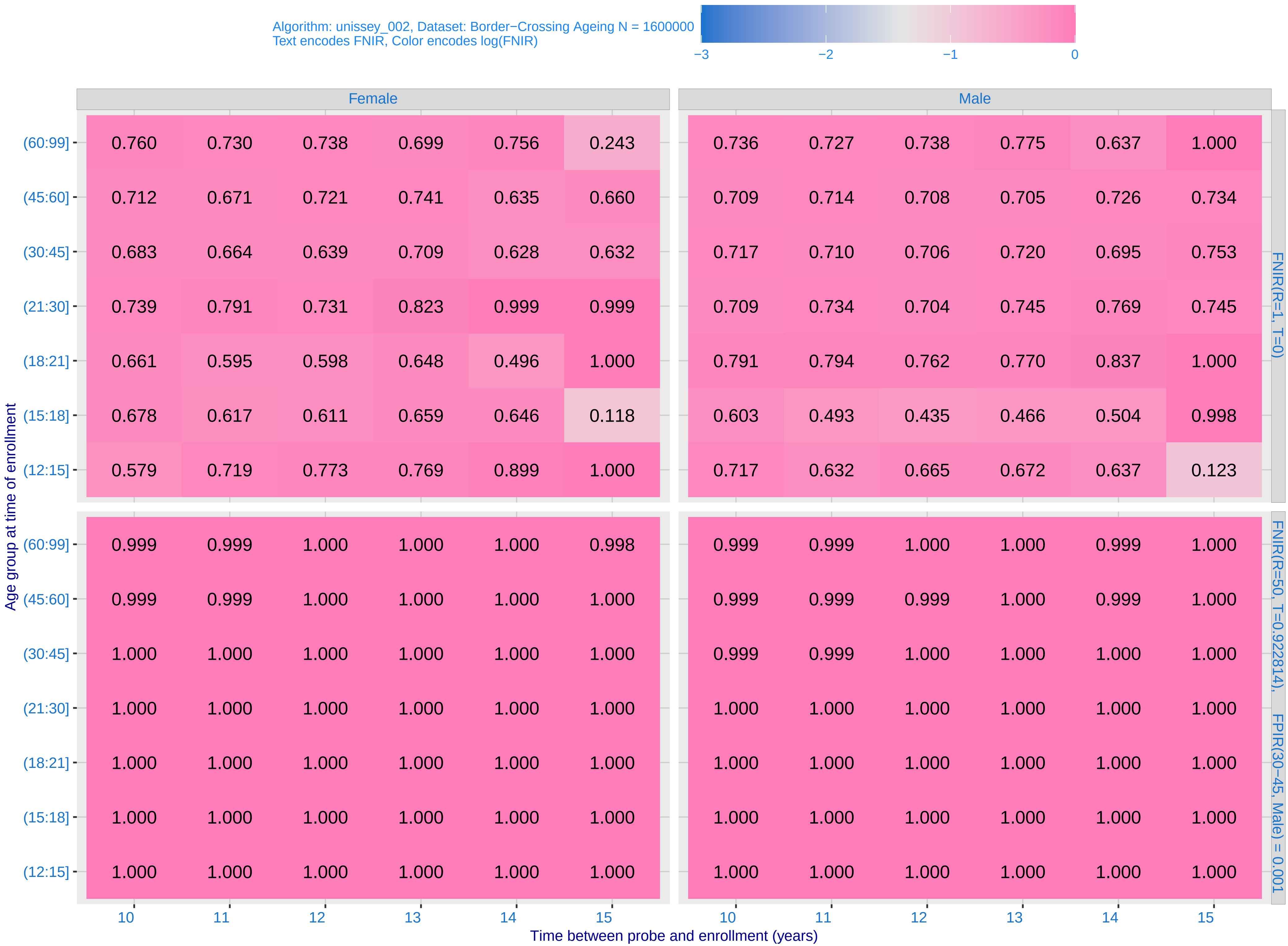


L: Investigational mode: FNIR(1600000, R, 0) by probe type
sensetime_009
unissey_002
0.700
0.500
0.300
0.200
0.100
0.070
0.050
0.030
0.020
0.010
0.007
0.005
0.003
0.002
0.001
False negative identification rate, FNIR(N)
1
3
10
30
Rank, R
enrolment_style
random
recent
FNIR(R)
N = 1600000
Immigration visa−border
Immigration visa−kiosk
Mugshot natural
Mugshot webcam

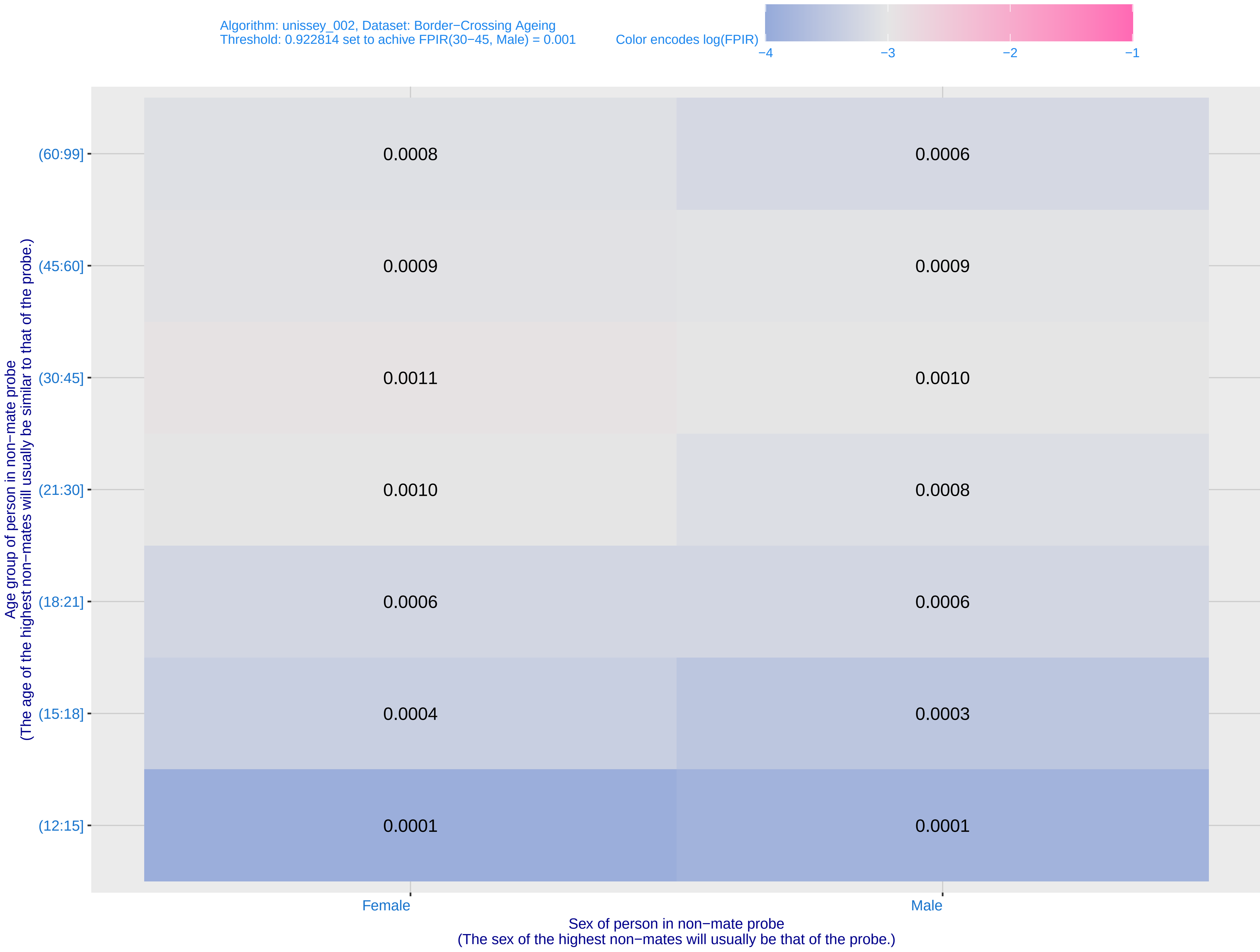
M: Template duration; search duration vs. N. The blue and pink ribbon covers 95 percent of observed measurements.
The template generation time is independent of N. The log and power−law models are fit to the first two (N,T) observations

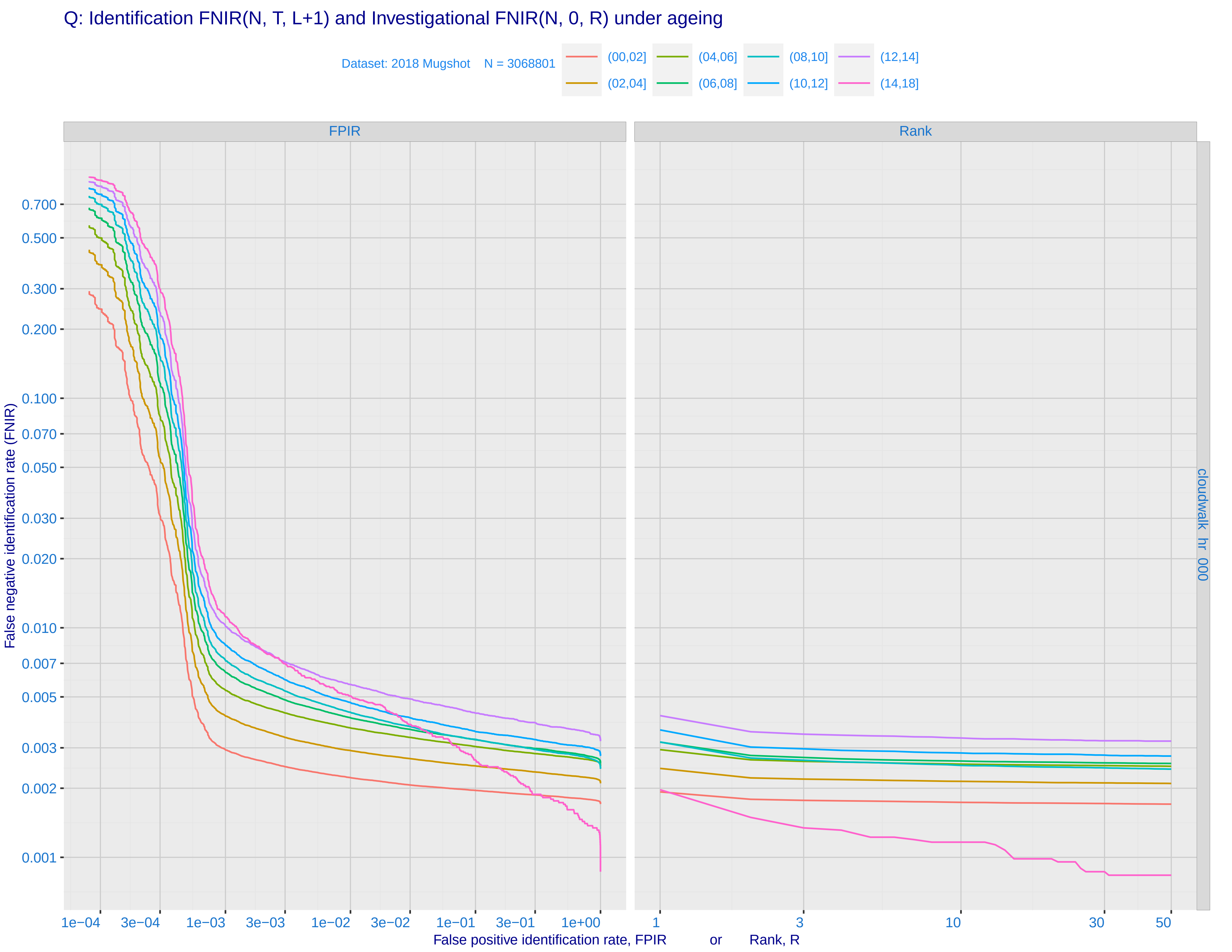


O: FNIR(T, N = 1.6 million) by sex, age and time−lapse. The top row gives investigational rank−1 miss rates.
The bottom panels give high threshold for more lights−out identification with low FPIR.



P: FPIR(N = 1.6 million) by sex and age. It is typical for false positive identification rates to be higher in women except in their teens.





Q: Identification FNIR(N, T, L+1) and Investigational FNIR(N, 0, R) under ageing
Dataset: 2018 Mugshot N = 3068801
(00,02]
(02,04]
(04,06]
(06,08]
(08,10]
(10,12]
(12,14]
(14,18]
FPIR
Rank
cloudwalk_hr_000
False negative identification rate (FNIR)
0.700
0.500
0.300
0.200
0.100
0.070
0.050
0.030
0.020
0.010
0.007
0.005
0.003
0.002
0.001
1e−04
3e−04
1e−03
3e−03
1e−02
3e−02
1e−01
3e−01
1e+00
1
3
10
30
50
False positive identification rate, FPIR or Rank, R